Lesson 14 Teacher's Key: Raising the Stakes

GameMaker - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Same idea as every engine this week - replace hardcoded literals with wave-dependent formulas - expressed as GML inside scr_start_game's spawn_wave() function, introduced in Lesson 11.

Level 1 - Full Worked Solution

- var count = 4 + wave; computes the count once, then reuses it for both the repeat loop and enemies_remaining - a single source of truth by construction.
- No new instance variables are needed for count scaling - only Level 2's speed scaling introduces one (enemy_speed on obj_enemy).
- Wave 1 still spawns exactly 5 (4 + 1 = 5), so nothing about the game's opening feel changes - only later waves escalate.

Why This Structure

- Storing count in a local var (rather than repeating the formula) guarantees the repeat loop and the reset value can never disagree - the same discipline as every other engine this week, in GML's own idiom.
- Setting enemy_speed on each freshly-created instance (inst.enemy_speed = ...) rather than changing obj_enemy's Create-event default keeps the scaling logic entirely inside spawn_wave() - obj_enemy's Step event stays exactly as simple as Lesson 7 left it, just reading a variable instead of a literal.

Level 2 / Level 3 Talking Points

- Ask students to trace spawn_wave() for wave 3 by hand: count = 4 + 3 = 7 - a good comprehension check.
- For students comparing engines: point out that TIC-80's identical (4 + wave) formula, expressed as plain Lua instead of GML, is conceptually the exact same idea - a good cross-engine parallel, and both engines this week needed to introduce a NEW per-instance/per-entry speed value where none existed before.

Common Student Mistakes

- Updating the repeat loop's count but forgetting to update enemies_remaining (or vice versa) - avoided entirely if count is stored in a local var and reused, which is why that's the recommended pattern.

- Setting enemy_speed on the wrong instance (e.g. self instead of inst, or vice versa) inside spawn_wave().
- Forgetting to change obj_enemy's Step event from "y += 1;" to "y += enemy_speed;" - the variable exists but is never actually read.